

Water recovery from saline water using hydrophobic polyvinylidene fluoride (PVDF) hollow fiber membrane based direct contact membrane distillation (DCMD)

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1 INTRODUCTION

- The working principle of MD relies on the vapor pressure difference of volatile substances between the upstream and downstream sides of the membrane, which serves as the driving force for separation.¹
- MD exclusively uses hydrophobic porous membranes because the separation phenomenon allows only vapor to pass through while preventing liquid water from transferring. To achieve this, only anti-wetting substances or polymers are utilized in membrane preparation.²
- This study focuses on direct-contact membrane distillation (DCMD) among four membrane distillation configurations, including vacuum membrane distillation (VMD), air-gap membrane distillation (AGMD), and sweeping-gas membrane distillation (SGMD).³
- DCMD was chosen for this study due to its simpler system design, lower energy consumption, improved heat recovery, and ease of scalability.

2 OBJECTIVE

- To fabricate a hydrophobic PVDF hollow fiber membrane with porous morphology**
 - designed to enhance vapor transport and minimize mass-transfer resistance during DCMD.
- To evaluate the desalination performance of the membrane in DCMD**
 - using a 3.5 wt% NaCl feed solution under a hot stream (80°C) and cold distillate stream (18°C), assessing permeation flux and salt rejection.
- To assess the long-term operational stability of the membrane**
 - by monitoring flux and rejection consistency over a continuous 10-hour DCMD run, confirming resistance to pore wetting.

3 METHODOLOGY

I Membrane preparation

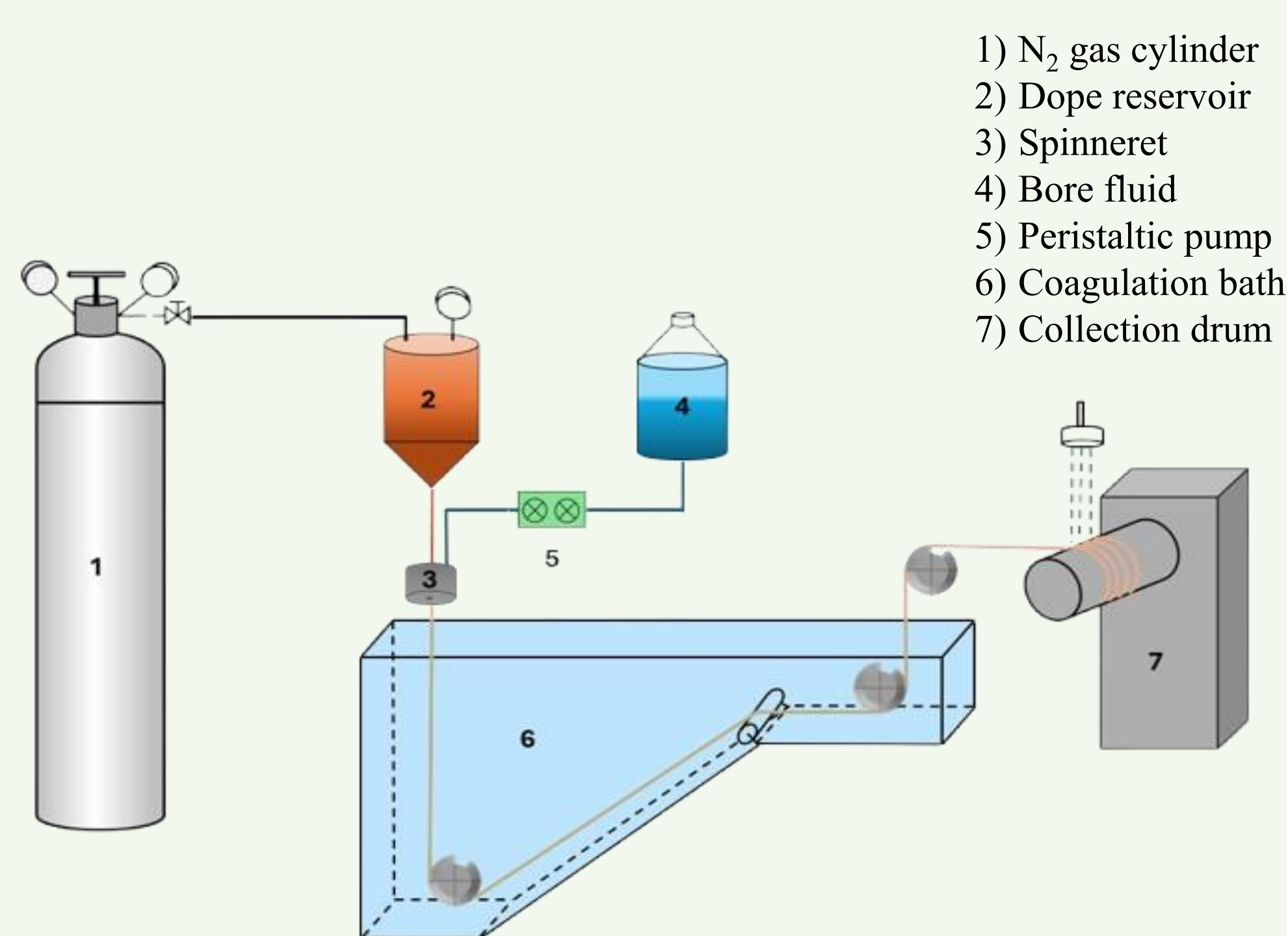


Fig. 1. Schematic of hollow fiber spinning unit.

II Testing setup

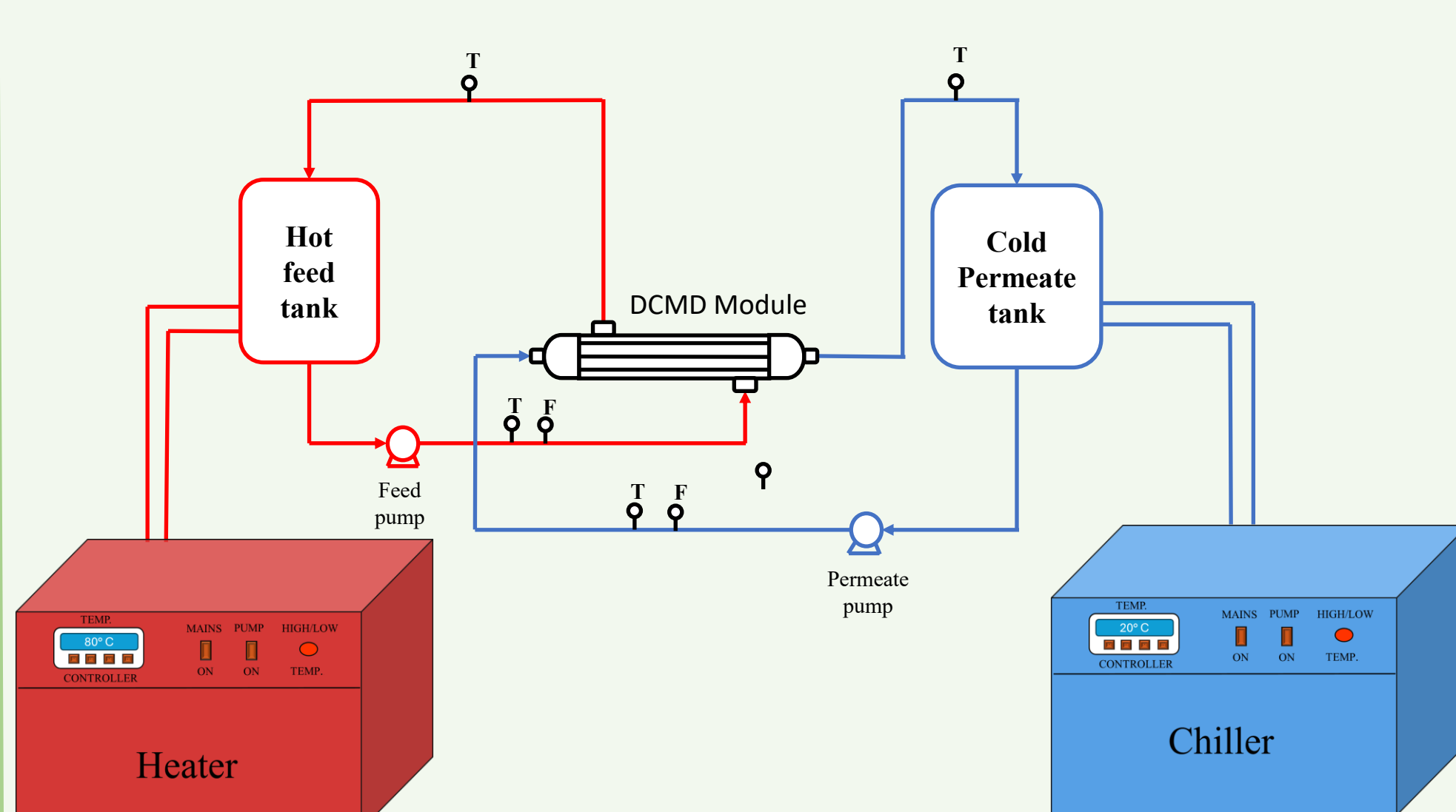


Fig. 2. Schematic of DCMD set-up.

4 CHARACTERIZATION

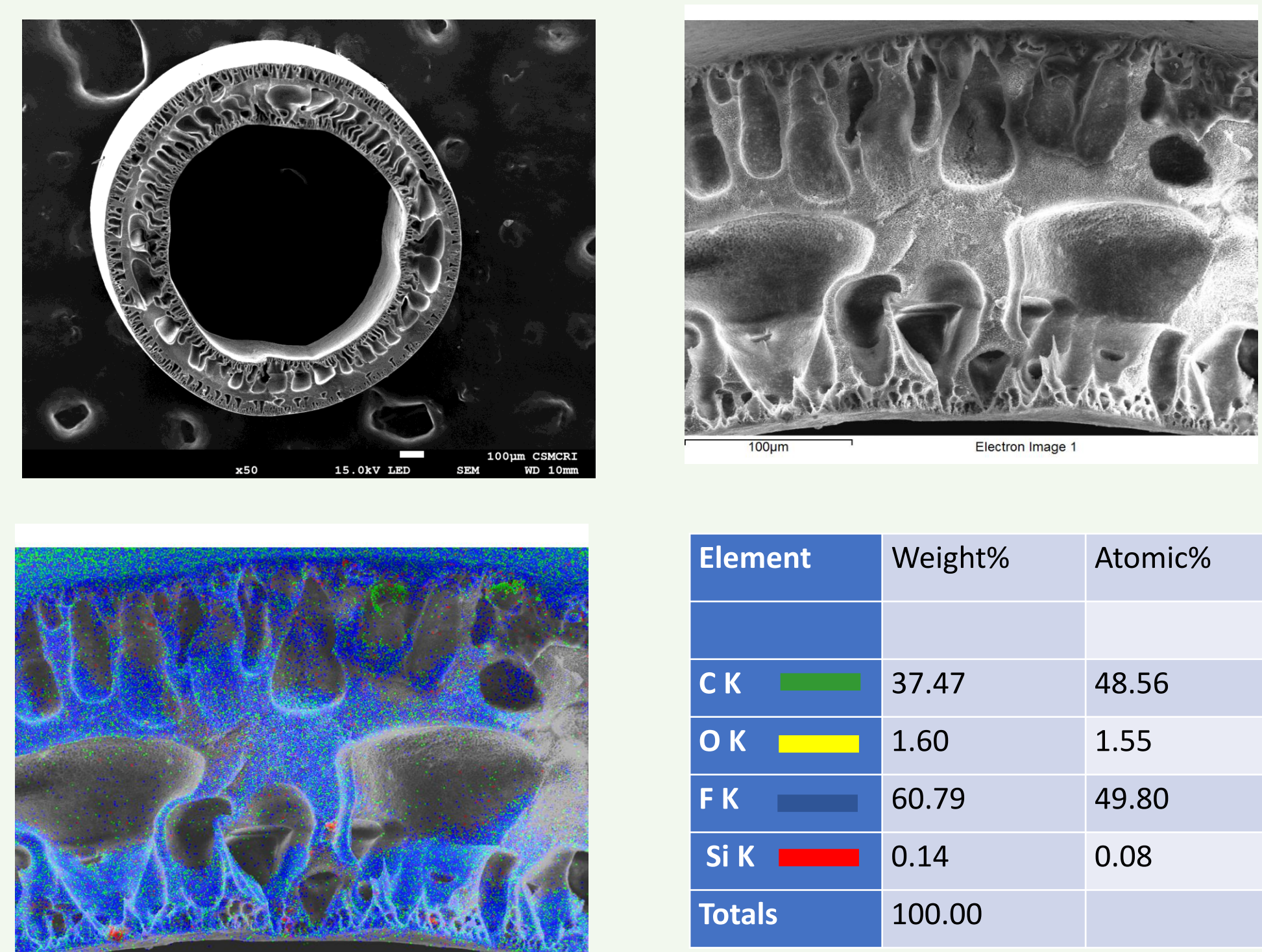


Fig. 3. SEM images, mapping and EDX of PVDF hollow fiber membrane

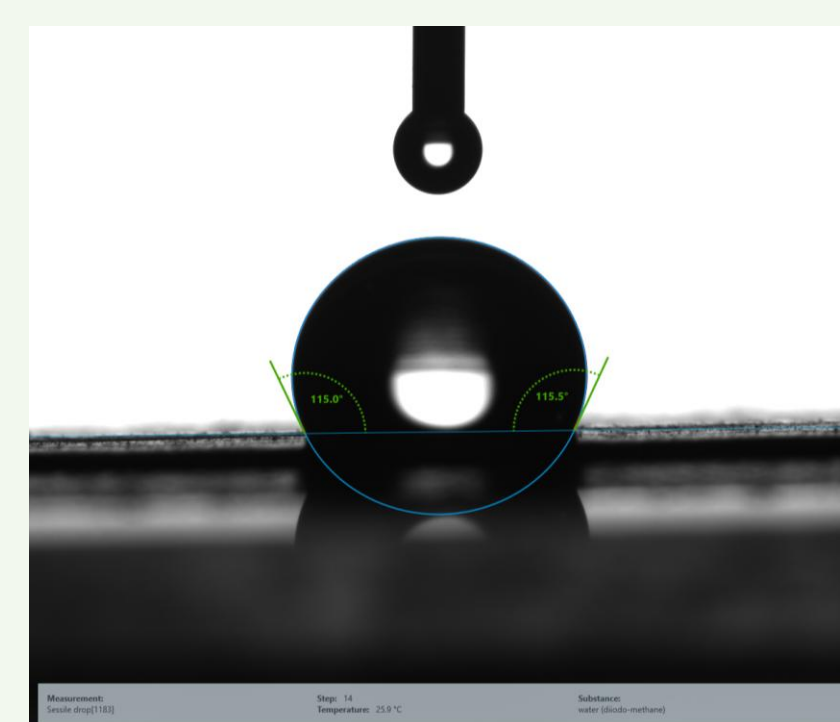
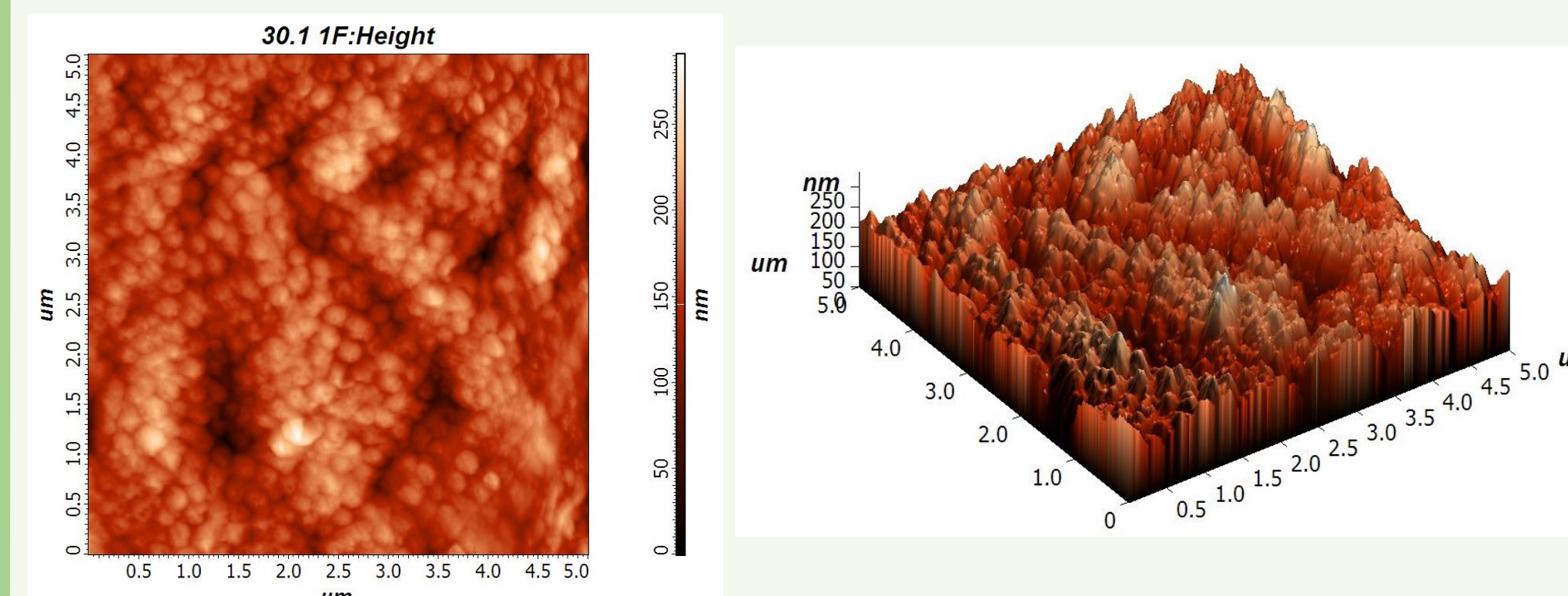


Fig. 4. Surface analysis by AFM and contact angle of hollow fiber membrane

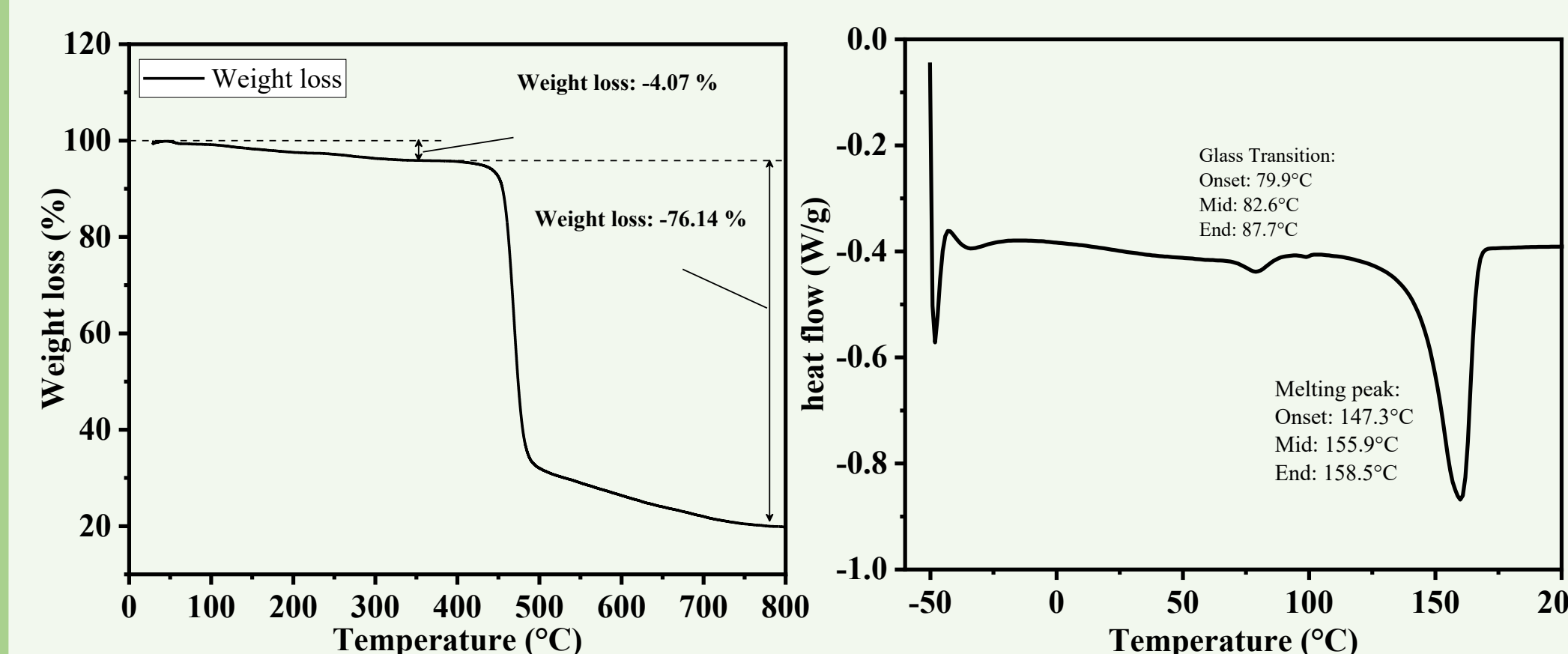


Fig. 5. Thermal analysis by TGA and DSC

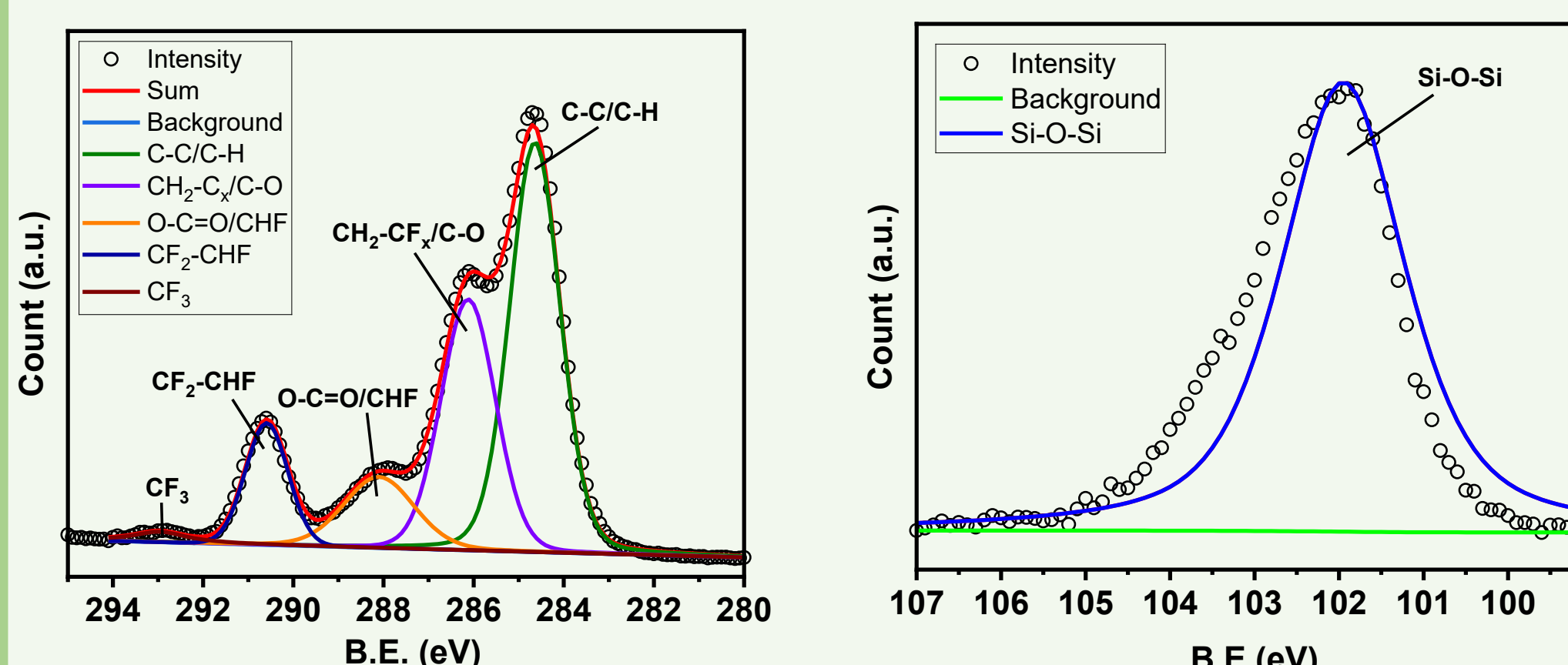


Fig. 6. XPS analysis of membrane.

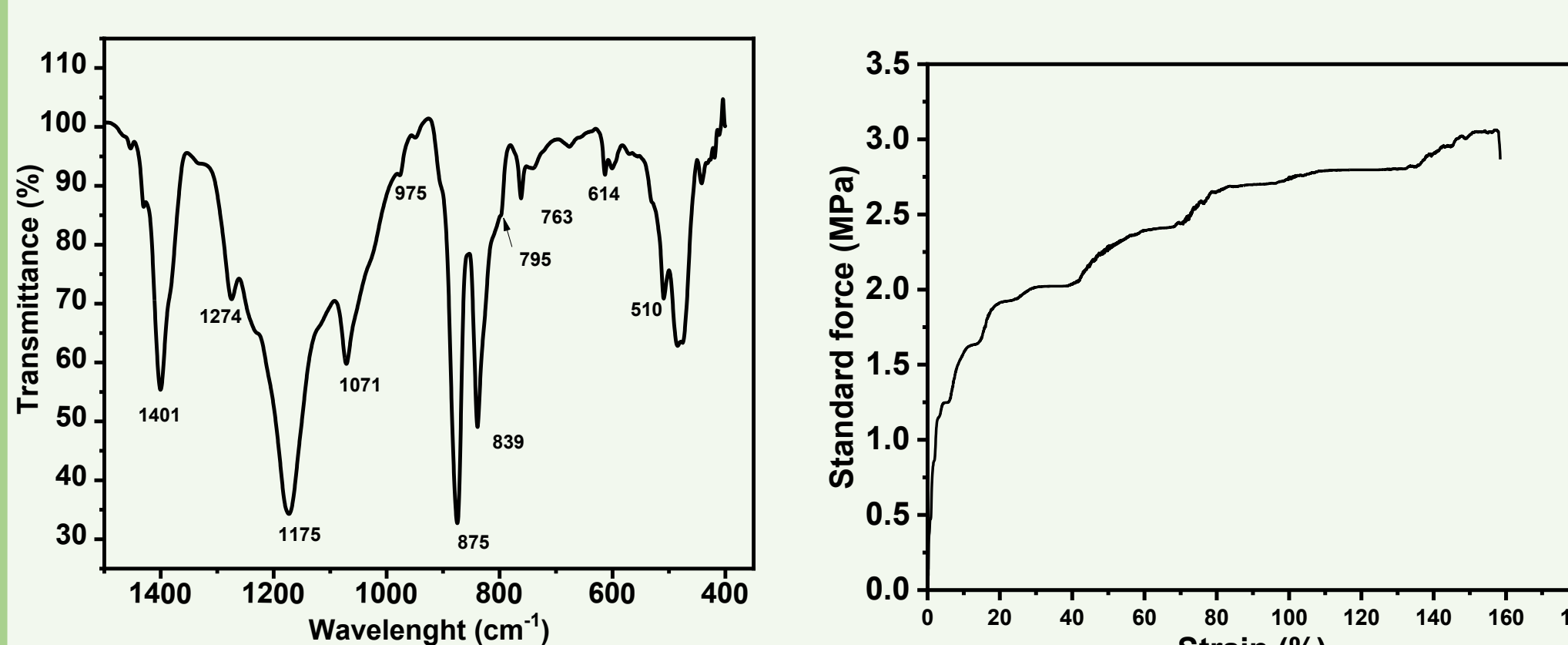
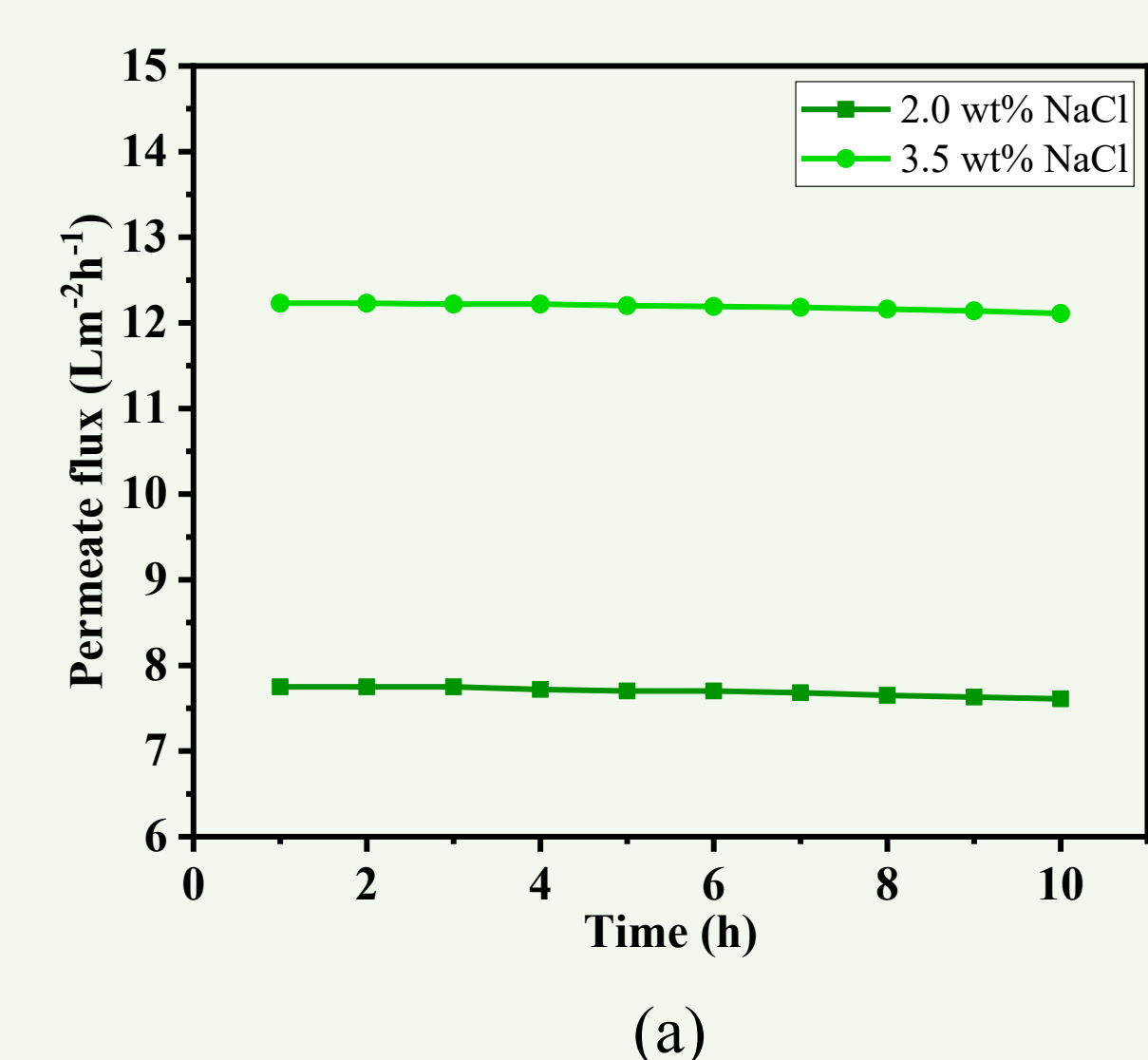
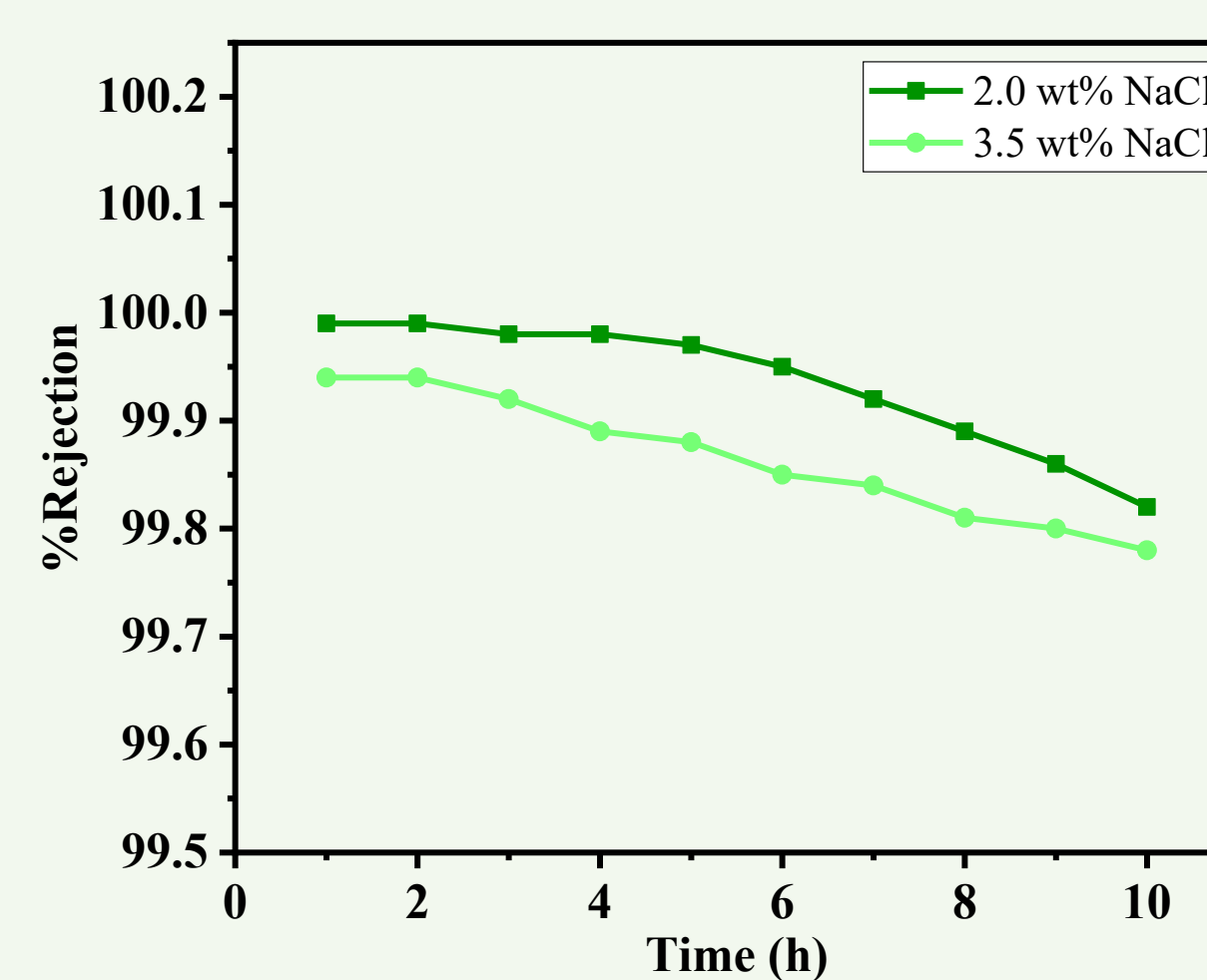


Fig. 7. Composition analysis by FTIR and tensile strength of membrane.

5 PERFORMANCE



(a)



(b)

Fig. 8. (a) Permeate flux, (b) NaCl rejection

6 CONCLUSION

- PVDF has inherently good thermal ability, chemical resistance and anti wetting property, Which is importance for DCMD.
- Feed temp. 80° and cold temp. 18° which is very low heat energy required for treatment of higher salt concentration.
- PVDF based hollow fiber membrane exhibits significant permeate vapor flux around 7.5 Lm⁻²h⁻¹ with exceeding 99.95% for 3.5 wt% NaCl.

7 FUTURE WORK

- Fabricating more hydrophobic membrane by adding modified nanoparticle like SiO₂ TiO₂ with TEOS or OTS.
- Post treatment like dip coating of hollow fiber membrane with hydrophobic PDMS.

8 REFERENCES

- [1] Pagliero, M., Khayet, M., García-Payo, C., & García-Fernández, L. (2021). Hollow fibre polymeric membranes for desalination by membrane distillation technology: A review of different morphological structures and key strategic improvements. *Desalination*, 516, 115235. <https://doi.org/10.1016/j.desal.2021.115235>
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- [3] K. Yu Wang, T.-S. Chung, M. Gryta, Hydrophobic PVDF hollow fiber membranes with narrow pore size distribution and ultra-thin skin for the fresh water production through membrane distillation, *Chemical Engineering Science* 63 (2008) 2587–2594.

9 ACKNOWLEDGE

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